

UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER EXAMINATION

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN

WEDNESDAY, 2 OCTOBER 2024

TIME : 2.00 PM – 4.00 PM (2 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS)
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
BUSINESS INFORMATION SYSTEMS
BACHELOR OF INFORMATION SYSTEMS (HONOURS)
INFORMATION SYSTEMS ENGINEERING
BACHELOR OF INFORMATION TECHNOLOGY (HONOURS)
COMPUTER ENGINEERING

Instruction to Candidates :

This question paper consists of **THREE (3)** questions in Section A and **TWO (2)** questions in Section B.

Answer **ALL** questions in Section A and **ONLY ONE (1)** question in Section B. Each question carries 25 marks.

Should a candidate answer more than **ONE (1)** question in Section B, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Candidates are allowed to use a calculator.

Answer questions only in the answer booklet provided.

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN

Section A (Compulsory Questions)

- Q1. (a) Identify **THREE (3)** types of requirement analysis technique used to analyse the user requirements. (3 marks)
- (b) Explain the differences between perfective maintenance and preventive maintenance in maintenance phase. (4 marks)

- (c) Based on Table 1 shown below:

ID	Activities	Predecessors	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)
A	Select technical staff	-	12	16	32
B	Select equipment	A	8	13	30
C	Interview applicants	A	4	14	18
D	Purchase equipment	B	19	32	33
E	Develop an information system	A	10	15	20
F	Install equipment	D	8	9	16
G	Staff training	C, E, F	2	9	10

Table 1: List of activities, their predecessors, and the projected completion times.

- (i) Calculate the expected task duration using program evaluation review technique (PERT). Given formula:
- $$TE = \frac{\text{Optimistic} + 4 \times \text{Most likely} + \text{Pessimistic time}}{6}$$
- where TE denoted as task expected time. (3 marks)
- (ii) Calculate the slack time for each task. (3 marks)
- (iii) Show the critical path in network diagram. (4 marks)
- (d) Discuss **TWO (2)** types of activities involved in closing down the project at the end of system development. (4 marks)
- (e) Identify **FOUR (4)** types of factors used to select an appropriate requirement determination technique to gather the information from user. (4 marks)

[Total : 25 marks]

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN

- Q2. (a) A user owns one or more vehicles. There are three types of vehicles: cars, motorcycles, and trucks. Each vehicle has a different engine capacity and number of wheels. Given the mentioned scenario, use the class diagram notation to illustrate the data storage structure of the system with the items listed below.
- (i) Association relationships between two classes. (3 marks)
 - (ii) Generalization relationships between two classes. (3 marks)
- [Hints: Combine the answer (i), and (ii) into a class diagram]
- (b) Identify and explain **THREE (3)** types of views used in architectural centric to show the system design from different perspectives. (9 marks)
- (c) Explain the differences between method and message in term of object-oriented concept. (4 marks)
- (d) Discuss **THREE (3)** types of method cohesion used to measure the quality of the design of methods. (6 marks)
- [Total : 25 marks]
- Q3. (a) ETS App is an application that allows users to book the train tickets online. First, the customer is required to login to his/her ETS App user account when he/she wants to book a train ticket. After that, the user can search for the availability of the train based on the travel date, source and destination to book a train ticket. Then, the user selects the train with an appropriate seat. The system will generate the fare price at the end of the train ticket booking. After the user makes the payment, the system will generate a copy of a ticket for the user. Given the mentioned scenario, use the context diagram notation to draw the items listed below.
- (i) Process. (1 mark)
 - (ii) Source and sink. (1 mark)
 - (iii) Data flows. (6 marks)
- [Hints: Combine the answer (i), (ii), and (iii) into a context diagram]

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Q3. (Continued)**

- (b) Given the flow of events,
- I. The consumer scans the identity card or passport.
 - II. The consumer selects the train with an appropriate seat.
 - III. The system checks fare price discount, if any. If the consumer is a senior citizen (aged 60 years or older) or a student, they will receive a 50% or 30% discount, respectively, on the fare price.
 - IV. The system generates the fare price.
 - V. The consumer makes the payment using a credit card or debit card.
 - VI. The system generates a copy of receipt for the consumer.

Based on the flow of events given, use the activity diagram with swimlane notation to draw the bus ticket purchasing process with items listed below.

- (i) Swimlane. (2 marks)
- (ii) Step-by-step activities of the bus ticket purchasing process. (4 marks)
- (iii) Decision nodes used for the bus ticket purchasing process. (4 marks)
- (iv) Merge node, control flows, initial node and final-activity node usage. (3 marks)

[Hints: Combine the answer (i), (ii), (iii) and (iv) into an activity diagram with swimlane]

- (c) Identify the **FOUR (4)** key components commonly used in a decision table. (4 marks)

[Total : 25 marks]

UCCD2003 OBJECT-ORIENTED SYSTEMS ANALYSIS AND DESIGN**Section B (Choose any ONE Question)**

- Q4. (a) Wallace is trying to decide which item to redeem or not redeem. The item can be redeemed using pure redemption points or redemption points with additional cash. She prefers to use pure redemption points for the item. However, if it is a household product, she is willing to use redemption points with additional cash. Additionally, she only redeems well-known brand items. Given the mentioned scenario, create a decision table. (10 marks)
- (b) Identify **THREE (3)** software architectural components used for physical architecture design. (3 marks)
- (c) Explain the differences between factoring, partitioning, and layering used in system design to achieve robustness, scalability, and maintainability across complex systems. (6 marks)
- (d) Discuss **THREE (3)** key components used in a deployment diagram. (6 marks)
- [Total : 25 marks]
- Q5. (a) Explain the differences among the layout, aesthetics, and minimal user efforts in terms of the principles of the user interface design. (6 marks)
- (b) List **FOUR (4)** ways of data storage optimization used in database design. (4 marks)
- (c) Users are required to log into their accounts on the Movie App to check movie ticket availability and make purchases. Additionally, they must display a QR code upon entering the movie theater. Given the mentioned scenario, draw a window navigation diagram. (8 marks)
- (d) Discuss the differences between the analysis model and design model in terms of user requirements. (4 marks)
- (e) Identify what is eliminated from 0NF to 1NF, from 1NF to 2NF, and from 2NF to 3NF, respectively, in terms of database normalization. (3 marks)
- [Note: NF represents normal form.]
- [Total : 25 marks]

